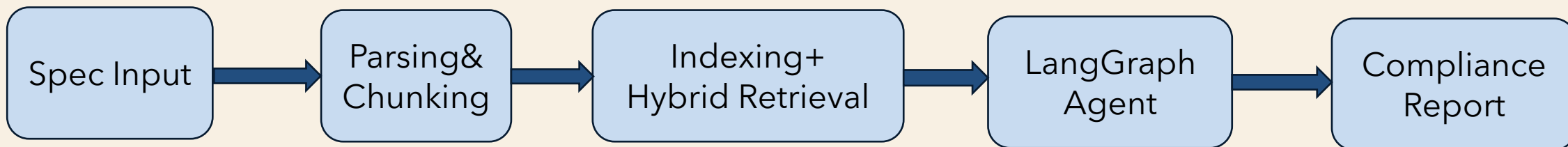


# IC SPEC GUARDIAN 文件審查 AGENT

IC Spec Guardian 是一個以 GenAI 為核心的自動化合規審查agent，目標是讓工程師在提交 Spec 之後，系統能夠混合檢索比對內部規範文件，找出潛在的違規點，並提出具體的修改建議。

系統流程：



# IC SPEC GUARDIAN:HYBRID RETRIEVAL

- **Chunking:**

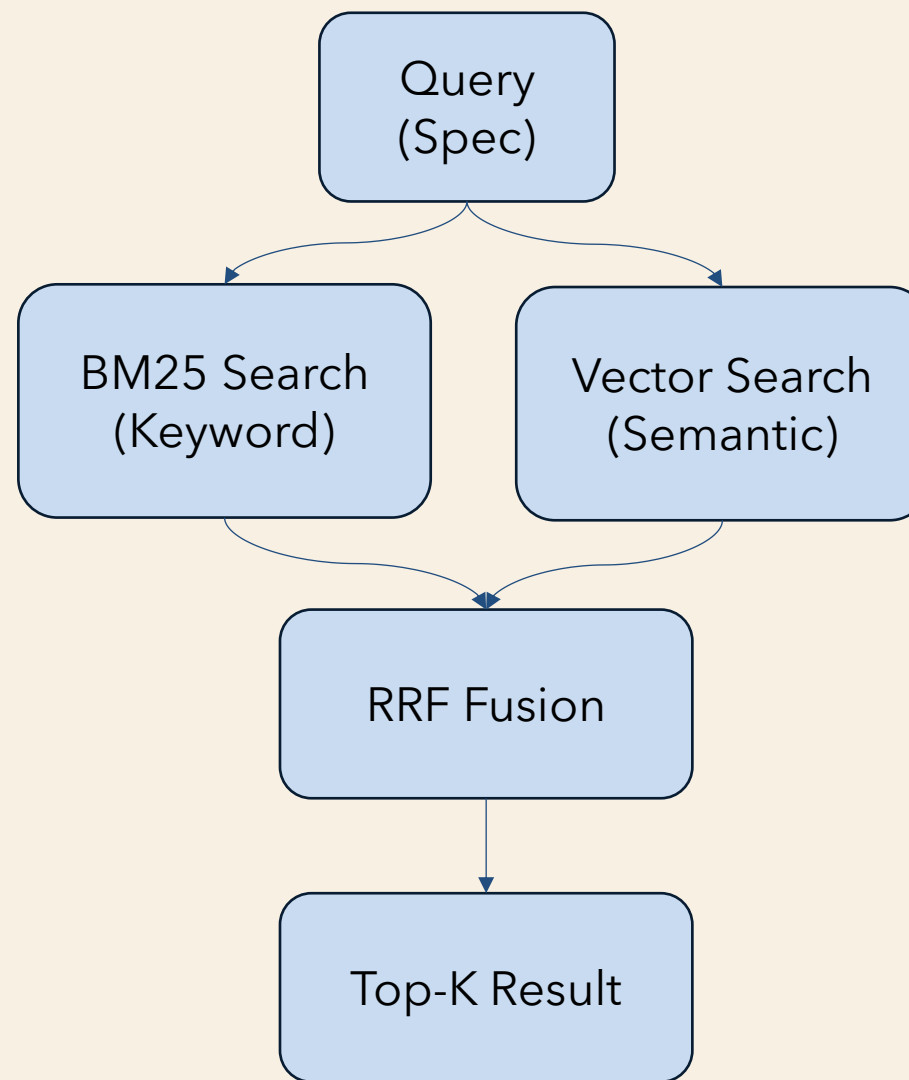
- 規則 chunk ( 有 rule\_id ) 保持完整
- 短文本不切
- 長且無結構文本做語意分割 semantic split

- **Indexing:**

- 1. BM25庫建立inverted index (keyword-based)
- 2. nomic-embed-text建立 Embedding
  - 存儲在Qdrant (vector database with ANN index)

- **Hybrid Retrieval**

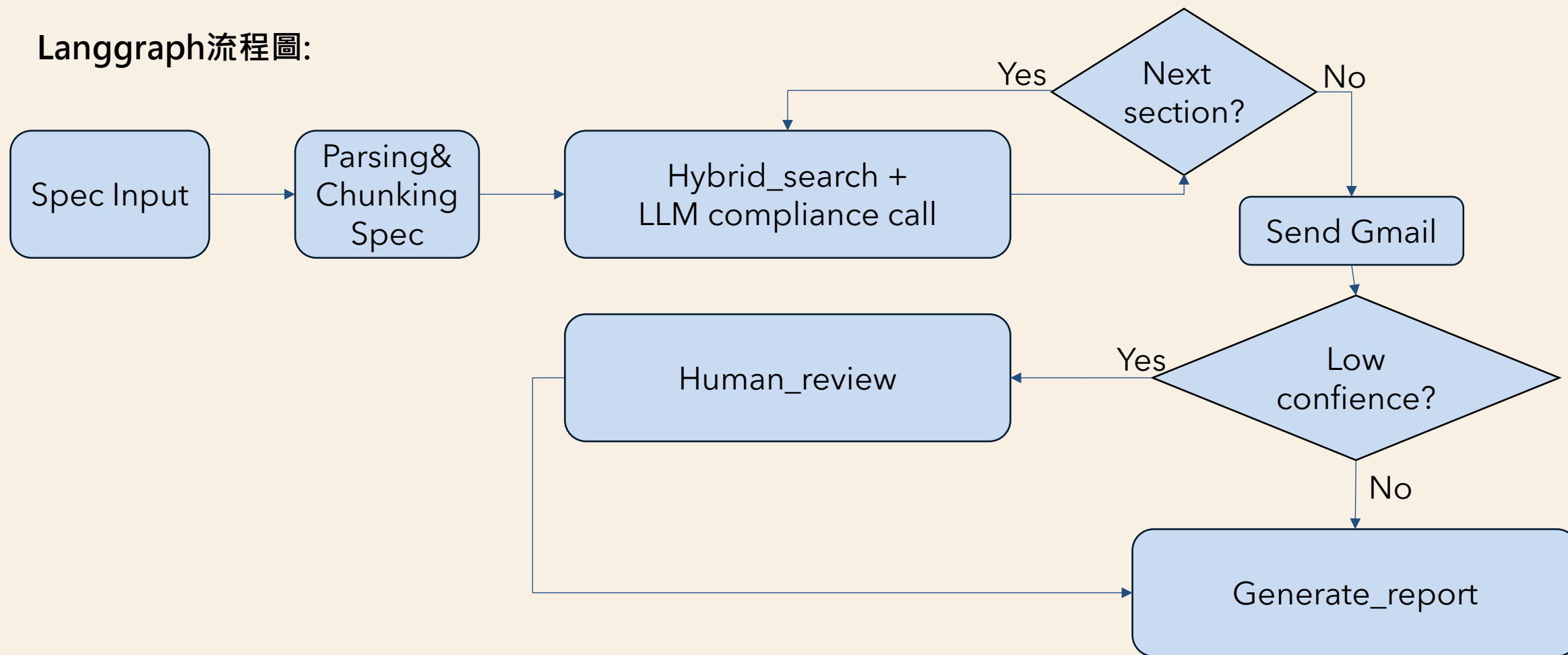
- 1. Sparse retrieval → BM25 (keyword match)
- 2. Dense retrieval → vector similarity search (Qdrant)
- 3. Fusion → Reciprocal Rank Fusion (RRF)
- 4. Top-K ranking



# IC SPEC GUARDIAN: LANGGRAPH

Highlight: Hallucination 控制(rule id check、confidence)、Checkpoint 與 Human-in-the-Loop

Langgraph流程圖:



# IC SPEC GUARDIAN: LANGFUSE

- 所有 LLM 呼叫都用 Langfuse 做追蹤，使用 @observe decorator 實現。每次審查會產生一個完整的 trace。
- 追蹤的關鍵數據：
  - 每段的 LLM 呼叫 latency
  - Input/Output /總token 用量
  - 自動計算的 Cost ( Langfuse 根據模型定價自動換算 )
  - 每次判斷的輸入 ( 章節長度、相關規則數 ) 和輸出 ( 是否違規、規則 ID、信心分數 )

# IC SPEC GUARDIAN: LANGFUSE

test Hobby / rag project

Tracing

Hide filters Views 4 Search... IDs / Names

|  | Start Time          | Type | Name                | Trace Name | Input |
|--|---------------------|------|---------------------|------------|-------|
|  | 2026-05-15 03:44:50 | ↔    | compliance_llm_call |            | ("ar  |
|  | 2026-05-15 03:44:49 | ↔    | compliance_llm_call |            | ("ar  |
|  | 2026-05-15 03:44:47 | ↔    | compliance_llm_call |            | ("ar  |
|  | 2026-05-15 03:42:42 | ↔    | compliance_llm_call |            | ("ar  |
|  | 2026-05-15 03:42:40 | ↔    | compliance_llm_call |            | ("ar  |
|  | 2026-05-15 03:42:39 | ↔    | compliance_llm_call |            | ("ar  |
|  | 2026-05-15 03:41:04 | ↔    | compliance_llm_call |            | ("ar  |
|  | 2026-05-15 03:41:03 | ↔    | compliance_llm_call |            | ("ar  |
|  | 2026-05-15 03:41:02 | ↔    | compliance_llm_call |            | ("ar  |
|  | 2026-05-15 03:39:18 | ↔    | compliance_llm_call |            | ("ar  |
|  | 2026-05-15 03:39:16 | ↔    | compliance_llm_call |            | ("ar  |
|  | 2026-05-15 03:39:14 | ↔    | compliance_llm_call |            | ("ar  |
|  | 2026-05-15 03:37:29 | ↔    | compliance_llm_call |            | ("ar  |
|  | 2026-05-15 03:37:27 | ↔    | compliance_llm_call |            | ("ar  |
|  | 2026-05-15 03:37:25 | ↔    | compliance_llm_call |            | ("ar  |
|  | 2026-05-15 03:36:12 | ↔    | compliance_llm_call |            | ("ar  |
|  | 2026-05-15 03:36:10 | ↔    | compliance_llm_call |            | ("ar  |
|  | 2026-05-15 03:36:06 | ↔    | compliance_llm_call |            | ("ar  |
|  | 2026-05-15 03:35:55 | ↔    | compliance_llm_call |            | ("ar  |
|  | 2026-05-15 03:35:53 | ↔    | compliance_llm_call |            | ("ar  |

Trace compliance\_llm\_call: 94309b3249cc5045be5fb75076fec7ef

Search Timeline

compliance\_llm\_call 1.15s 1,241 → 150 (Σ 1,391) \$0.000747

compliance\_llm\_call :  
+ Add to datasets Annotate Add comment

2026-05-15 03:44:50.761  
Latency: 1.15s Env: default \$0.000747 gemini-2.5-flash

Preview Scores Log View Formatted JSON Beta

Input

```
{ 2 Items
  args: [ 2 Items
    0: "System initialization follows a hardware-managed sequence with dynamic software adjustments.

- **Reset Ordering**: Power-on reset release follows the standard sequence. However, the CPU reset may be released **2 cycles after peripheral ready assertion** to overlap cache initialization, reducing boot time by 15%.
- **Software Reset Handshake**: Software-triggered reset for the display controller uses a **dynamic delay mechanism**. Instead of polling a hardware flag, the driver calculates a wait

...expand (496 more characters)
"
    1: [ 3 Items
      0: { 7 Items
        id: 12
        score: 0.73660827
        text: "## RST-001 Reset Sequence Ordering

**類別**: Reset
**嚴重程度**: Critical

**規定**:
多個 IP 的 reset 釋放必須遵循以下強制順序：
1. Clock 穩定 (PLL lock)
2. Memory (SRAM / register file) 初始化完成
3. Peripheral IP reset 釋放
4. Master IP (CPU / DMA) reset 釋放

Master IP 的 reset 不得在 Peripheral IP 完成 reset 之前釋放。

**違規範例**:
"The CPU reset and peripheral reset are released simultaneously after pll lock "
```

# IC SPEC GUARDIAN: 品質保證

## 1. RAGAS : 用 20 題針對規範文件撰寫的 QA 對作為 ground truth

- context\_precision : 0.8667 , 檢索回來的 chunks 有多少是真正相關的 ( 精準度 )
- context\_recall : 1.0 , 真正相關的規則有多少被檢索回來 ( 召回率 )
- faithfulness : 0.9131 , LLM 的答案有多忠實於檢索到的內容 ( 不 hallucinate )
- answer\_relevancy : 0.8057 , 答案與問題的相關程度

## • 2. Agent 評測 ( 推理品質 ) : Agent 評測針對合規審查任務本身 , 用 violation\_spec.md ( 含 3 個明確違規 ) 和 compliant\_spec.md ( 零違規 ) 作為測試集

- Precision:1 , Agent 宣稱的違規中 , 真正是違規的比例
- Recall:1 , 實際存在的違規中 , 被 Agent 找到的比例
- F1: 1

# IC SPEC GUARDIAN

IC SPEC // GUARDIAN Connected

SPEC FILE

[DOC]

Click to select file

Supports .md / .txt / .pdf

[F] ambiguous\_spec.md

3.3 KB

X

THREAD ID

review-1778794889676

API ENDPOINT

http://localhost:8082

START REVIEW

STATISTICS

1

High-conf violations

2

Pending review

0

Confirmed

0

Rejected

ISSUES 2

VIOLATIONS 1

REPORT

LOG

PENDING

AXI-001 Conf 78%

S. 1. AXI Interface Specification

AXI4 Master 介面聲稱支援 INCR 和 WRAP burst 模式，但對於高吞吐影片編碼路徑，WRAP 模式被韌體透過 runtime toggle 禁用。這違反了規則 AXI-001 的規定，即所有 AXI4 Master 介面必須同時支援 INCR 和 WRAP 兩種 burst 模式。

Suggest:

確保 AXI4 Master 介面始終同時支援 INCR 和 WRAP burst 模式，即使在特定路徑下需要禁用，也應確保其在硬體層面是可用的，並由軟體正確配置。或者，如果禁用 WRAP 模式是為了效能或設計上的考量，需要提供額外的驗證文件來證明此行為不會影響 legacy cache IP 的相容性。

VIOLATION

COMPLIANT

PENDING

CLK-001 Conf 70%

S. 2. Clock Domain Crossing Architecture

The 'fast interrupt request line ('irq\_fast')' uses a single FF stage followed by an asynchronous reset flip-flop to meet a <2ns latency budget. Rule CLK-001 requires at least two D flip-flop synchronizer stages for single-bit signals crossing clock domains.

Suggest:

Implement a standard 2-stage DFF synchronizer for the 'fast interrupt request line ('irq\_fast')' clocked by the destination domain, or provide a robust justification and verification plan for the single-stage synchronizer and asynchronous reset flip-flop combination that demonstrates it meets the metastability requirements.

VIOLATION

COMPLIANT

Please review 2 pending items

SUBMIT DECISIONS

7

# IC SPEC GUARDIAN

IC SPEC // GUARDIAN

Connected

SPEC FILE

[DOC]

Click to select file

Supports .md / .txt / .pdf

[F]

ambiguous\_spec.md

3.3 KB

X

THREAD ID

review-1778794582182

API ENDPOINT

http://localhost:8002

START REVIEW

Submitting decisions...

STATISTICS

1

High-conf violations

2

Pending review

1

Confirmed

1

Rejected

ISSUES 2

VIOLATIONS 1

REPORT

LOG

VIOLATION

AXI-001 Conf 78

S. 1. AXI Interface Specification

AXI4 Master 介面聲稱支援 INCR 和 WRAP burst 模式，但對於高吞吐量影片編碼路徑，WRAP 模式被韌體透過 runtime toggle 禁用。這違反了規則 AXI-001 的規定，即所有 AXI4 Master 介面必須同時支援 INCR 和 WRAP 兩種 burst 模式。

Suggest:

確保 AXI4 Master 介面始終同時支援 INCR 和 WRAP burst 模式，即使在特定路徑下需要禁用，也應確保其在硬體層面是可用的，並由軟體正確配置。或者，如果禁用 WRAP 模式是為了效能或設計上的考量，需要提供額外的驗證文件來證明此行為不會影響 legacy cache IP 的相容性。

COMPLIANT

CLK-001 Conf 78

S. 2. Clock Domain Crossing Architecture

The 'fast interrupt request line ('irq\_fast')' uses a single FF stage followed by an asynchronous reset flip-flop to meet a <2ns latency budget. Rule CLK-001 requires at least two D flip-flop synchronizer stages for single-bit signals crossing clock domains.

Suggest:

Implement a standard 2-stage DFF synchronizer for the 'fast interrupt request line ('irq\_fast')' clocked by the destination domain, or provide a robust justification and verification plan for the single-stage synchronizer and asynchronous reset flip-flop combination that demonstrates it meets the metastability requirements.

8



# IC SPEC GUARDIAN

IC SPEC // GUARDIAN Connected

SPEC FILE

[DOC]

Click to select file

Supports .md / .txt / .pdf

[F]

ambiguous\_spec.md

3.3 KB

X

THREAD ID

review-1778787887516

API ENDPOINT

http://localhost:8002

START REVIEW

Submitting decisions...

STATISTICS

1

High-conf violations

2

Pending review

2

Confirmed

0

Rejected

ISSUES 2

VIOLATIONS 1

REPORT

LOG

HIGH CONF VIOLATION

RST-001 Conf 80%

S. 3. Power & Reset Sequence

CPU reset may be released 2 cycles after peripheral ready assertion to overlap cache initialization, reducing boot time by 15%. This violates the rule that Master IP reset must not be released before Peripheral IP completes reset.

Suggest:

Ensure that the CPU reset is released only after all peripheral IPs have completed their reset sequence and asserted their ready signals. The sequence should be: PLL lock → peripheral reset release → peripheral ready → CPU reset release.

# IC SPEC GUARDIAN

IC SPEC // GUARDIAN Connected

SPEC FILE

[DOC]

Click to select file

Supports .md / .txt / .pdf

[F] ambiguous\_spec.md

3.3 KB

X

THREAD ID

review-1778787887516

API ENDPOINT

http://localhost:8082

START REVIEW

Submitting decisions...

STATISTICS

1

High-conf violations

2

Pending review

2

Confirmed

0

Rejected

ISSUES 2

VIOLATIONS 1

REPORT

LOG

⚠️ ambiguous\_spec.md 審查報告

發現 3 個問題

=====

[問題 1] 人工確認

章節: 1. AXI Interface Specification

違反規則: AXI-001

描述: AXI4 Master 介面聲稱支援 INCR 和 WRAP burst 模式，但對於高吞吐量影片編碼路徑，WRAP 模式被韌體透過 runtime toggle 禁用。這違反了規則 AXI-001 的規定，即所有 AXI4 Master 介面必須同時支援 INCR 和 WRAP 兩種 burst 模式。

建議: 確保 AXI4 Master 介面始終同時支援 INCR 和 WRAP burst 模式，即使在特定路徑下需要禁用，也應確保其在硬體層面是可用的，並由軟體正確配置。或者，如果禁用 WRAP 模式是為了效能或設計上的考量，需要提供額外的驗證文件來證明此行為不會影響 legacy cache IP 的相容性。

信心分數: 0.70

工程師備注: Engineer confirmed

[問題 2] 人工確認

章節: 2. Clock Domain Crossing Architecture

違反規則: CLK-001

描述: The 'fast interrupt request line ('irq\_fast')' uses a single FF stage followed by an asynchronous reset flip-flop to meet a <2ns latency budget. Rule CLK-001 requires at least two D flip-flop synchronizer stages for single-bit signals crossing clock domains.

建議: Implement a standard 2-stage DFF synchronizer for the 'fast interrupt request line ('irq\_fast')' clocked by the destination domain, or provide a robust justification and verification plan for the single-stage synchronizer and asynchronous reset flip-flop combination that demonstrates it meets the metastability requirements.

信心分數: 0.70

工程師備注: Engineer confirmed

[問題 3] 高信心違規

章節: 3. Power & Reset Sequence

違反規則: RST-001

描述: CPU reset may be released 2 cycles after peripheral ready assertion to overlap cache initialization, reducing boot time by 15%. This violates the rule that Master IP reset must not be released before Peripheral IP completes reset.

建議: Ensure that the CPU reset is released only after all peripheral IPs have completed their reset sequence and asserted their ready signals. The sequence should be: PLL lock → peripheral reset release → peripheral ready → CPU reset release.

信心分數: 0.80

DOWNLOAD REPORT

10